Fiscal Year:	FY 2021	**Task Last Updated:**	FY 05/11/2022
PI Name:	Morrell, Gerardo Ph.D.		
Project Title:	Causal Multivariate Network Analysis of Multi-Omics Datasets for Therapeutic Treatment of Muscle Atrophy in Mice and Homosapiens in Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	00936	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	EPSCoR-Space Biology NNH20ZHA001C
Start Date:	06/01/2020	**End Date:**	05/31/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA EPSCoR
Contact Monitor:	Compton, Jeppie	**Contact Phone:**	
Contact Email:	jeppie.r.compton@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/31/2022 (original end date was 5/31/2021) per NSSC information (Ed., 1/20/22)		
Key Personnel Changes/Previous PI:	Ed. Note: Heeralal Janwa, Ph.D. was added as a CoI for expertise in network analysis (5/11/22). Dr. Gerardo Morell serves as the Puerto Rico NASA EPSCoR Director, he/she is the administrative Principal Investigator on this proposal and will oversee the overall progress of the work and report to NASA.		
COI Name (Institution):	Cekanaviciute, Egle Ph.D. (NASA Ames Research Center) Manian, Vidya Ph.D. (University of Puerto Rico) Janwa, Heeralal (University of Puerto Rico)		
Grant/Contract No.:	80NSSC20M0132		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Established Program to Stimulate Competitive Research (EPSCoR) Project The goal of this proposal is to perform multi-study omics data integration of mice NASA GeneLab datasets from the International Space Station (ISS) and predict muscle atrophy in microgravity, beneficial for diagnosis and therapeutic intervention for humans (Homo sapiens) in spaceflight. Muscle atrophy is the wastage of muscle tissue that occurs due to aging, genetics, degenerative diseases, and injuries. This condition is common among astronauts due to exposure to microgravity. Understanding the effects of novel therapies to treat muscle atrophy in mice is essential to provide similar treatments to human astronauts, as well as patients affected by this condition on Earth. Our study will focus on the analysis of Gene Regulatory Networks (GRNs) of mouse in spaceflight and on Earth. Our current study has shown to be successful with the plant Arabidopsis thaliana gene expression GeneLab Data System (GLDS) GLDS7, and GLDS120 datasets. Using graph based GRNs we will identify key gene players from these datasets, and identify gene signaling pathways that are associated with muscle and tissue degradation, ultimately leading us to GRN based therapeutics in spaceflight. Muscle atrophy and its treatments and drug-designs have been well studied on Earth, using GRNs, hence we will find correlates between the networks constructed on ground, with those on spaceflight. We will determine causal relations between the effects of therapy on ground, and on spaceflight using multivariate analysis. The objectives for NASA Rapid Response Project are: 1) Analysis from flight and ground: Determine gene regulatory pathways, and common pathways in these networks using Fisher's analysis. 2) Construct causal relational networks for muscle atrophy and its treatment, and implement probabilistic deep learning networks to predict the best therapies for muscle atrophy in spaceflight. The GeneLab datasets (GLDS-250, 249, 246, 245, 247, 248, 243, 244) and ground based Gene Expression Omnibus (GEO) datasets (https://) on muscle atrophy and the effects of therapy will be used to identify top most common pathways. A causal relational network analysis will be performed to predict the best treatment and countermeasures for muscle atrophy in microgravity. The causal networks and the results of analysis generated by this project will be made available to the space biology research community. Dr. Nathaniel Szewczyk (Professor of Nottingham Biomedical Research Centre) and Dr. Jonathan Galazka (Scientist from NASA Space Biosciences Research Branch) will be collaborating in this project. They will be providing feedback and insights in to the progress of this research activity. Their backgrounds in medicine and biology will be indispensable for this project. This project will be conducted at the University of Puerto Rico (UPR), Mayaguez Campus and Rio Piedras Campus. The project team is interdisciplinary bringing together Dr. Manian, a faculty in Electrical & Computer Engineering, and Bioengineering and Dr. Janwa, a faculty in Mathematics. This project is conducted in a Hispanic minority serving institution with 99% Hispanic students who will gain knowledge in the fields of space biology, mathematics, bioinformatics, and software tool development in Python, R, SageMath, and Cython (with resulting packages freely available for other scientists). This project will directly support two graduate students in Space Biology research, and involve many undergraduate students through course work and workshops. They will be exposed to NASA space related research. This project will also enable Hispanic minority students to take up internships with NASA Labs and add to the workforce development for future NASA missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the effects of novel therapies to treat muscle atrophy in mice is essential to provide similar treatments to human astronauts, as well as patients affected by this condition on Earth.
Task Progress:	We have used the Pearson correlation method and Bayesian network methods for generating gene regulatory networks from the muscle atrophy gene expression datasets available in GeneLab. We have applied Fisher analysis to determine the key genes from each of these datasets. There are about 400 genes affected in the spleen, liver, and thymus of mice flown in spaceflight microgravity. These genes are shared in the GLDS-4, 244, 245, 246, 288, and 289 datasets. We have also found that these genes are differentially regulated in many common diseases that affect humans on Earth such as cancer, diseases of cellular proliferation, cognitive disorder, respiratory diseases, urinary tract diseases, intestinal disorders, muscle disorders, dermatitis, cardiovascular hypertension, bone diseases, and diabetes. We found that the genes that show higher activations due to muscle atrophy in spaceflight are the ones that are affected due to the diseases listed above on ground. This indicates that muscle atrophy is a consequence or side-effect of several diseases prevalent on Earth or it could be an indicator or forewarning of diseases that may occur in subjects who have a higher expression of these genes in spaceflight or ground. This project has substantially expanded research in graph based deep learning methods, and network sciences and applications in Puerto Rico. This research has resulted in research collaborations among several campuses of the University of Puerto Rico (UPR) system (three of them major research centers). Novel courses such as Network Analysis and Applications (UPR Rio Piedras), Pattern Recognition and Machine Learning and Applications (at UPR Mayaguez), have been offered that have trained a much larger body of graduate students and researchers. This project has enabled the collaboration of faculty and students between the UPR-Mayaguez and UPR-Rio Piedras campuses, specifically in the departments of Electrical Engineering, Bioengineering, and Mathematics. Our research has resulted in two proposals that have been submitted that expand related research that is being carried out. We are currently writing two journal publications that will include our current and future results from this project. We have done related conference presentations. [Ed. Note: The manuscripts under preparation have now been published and are listed in the bibliography for this report (5/11/22).]
Bibliography Type:	Description: (Last Updated: 03/03/2023)
Articles in Peer-reviewed Journals	Manian V, Orozco-Sandival J, Diaz-Martinez V. "An integrative network science and artificial intelligence drug repurposing approach for muscle atrophy in spaceflight microgravity. " Front Cell Dev Biol. 2021 Sep 16;9:732370. https://doi.org/10.3389/fcell.2021.732370 ; PMID: 34604234; PMCID: PMC8481783, Sep-2021 , Sep-2021

Articles in Peer-reviewed Journals	Manian V, Orozco-Sandival J, Diaz-Martinez V, Janwa H, Agrinsoni C. "Detection of target genes for drug repurposing to treat skeletal muscle atrophy in mice flown in spaceflight." Genes. 2022 Mar 8;13(3):473. https://doi.org/10.3390/genes13030473 PMID: 35328027; PMCID: PMC8953707 , Mar-2022